Related Work:

Alzheimer’s disease (AD) gradually breaks down memory, cognitive skills, and a person's ability to handle daily tasks. Catching it early makes a massive difference in how we manage the disease and plan treatments. Lately, Magnetic Resonance Imaging (MRI) has been the go-to tool for spotting the physical changes AD causes in the brain, like cortical thinning and hippocampal atrophy. Alongside this, deep learning has really stepped up, giving us much more accurate and automated ways to diagnose the condition.

When researchers first started using deep learning to classify Alzheimer's, they mostly leaned on Convolutional Neural Networks (CNNs). This approach was highly intuitive, as CNNs are naturally good at picking out spatial features from MRI scans. For example, Wen et al. showed that CNNs do a solid job classifying AD from structural MRIs. However, they also raised valid concerns about how consistent and reproducible medical imaging studies actually are. El-Assy et al. also built CNN architectures specifically for catching AD early on, proving that deep learning easily beats out older machine learning methods when it comes to pulling out features and classifying scans accurately.

But even with these successes, basic CNNs hit a wall when dealing with highly complex medical data. They often lack the depth needed for rich feature representation, which naturally pushed the field toward deeper, more efficient network designs.

To get past the limits of standard CNNs, a lot of research shifted toward transfer learning. Models like VGG16, VGG19, DenseNet, Xception, and InceptionResNetV2 have done really well in medical imaging because they don't start from scratch—they borrow pre-trained knowledge from massive datasets like ImageNet.

Mousavi et al., for instance, found that models trained on big MRI datasets could accurately classify Alzheimer's across its different stages. We've also seen that combining models—like putting VGG16, MobileNet, and InceptionResNetV2 together in an ensemble—makes the diagnosis much more robust. Looking at specific architectures, DenseNet boosts performance by linking layers densely, which helps gradients flow better and reuses features. Xception takes a different route, using depthwise separable convolutions to save on computing power without losing accuracy. Then there's InceptionResNetV2, which mixes multi-scale feature extraction with residual learning—a combination that handles the complexity of MRI scans exceptionally well.

Recently, the focus has shifted heavily toward ensemble learning and comparing different deep learning setups side-by-side. By combining various CNN models, researchers have noticed better stability and accuracy. Papers in journals like *PLOS ONE* point out that mixing architectures helps the models generalize much better across different kinds of MRI data.

Interestingly, comparative studies show us that there isn't one "perfect" architecture that wins every time on every dataset. This tells us how crucial it is to test multiple models under the exact same experimental conditions. Because of this, the trend is moving away from just building single models and toward systematic, side-by-side performance comparisons.

If you look at systematic reviews in this field, it's clear deep learning has changed the game for AD detection via MRI. But we're still facing some real hurdles. Things like imbalanced datasets, a lack of external validation, and the tricky issue of data leakage when using augmented images are still major problems. On top of that, a lot of papers just chase high accuracy scores and ignore whether doctors can actually understand or use the model in a clinic.

Recent surveys keep stressing that we need models that are lightweight, explainable, and ready for the real world. So, while we've made great progress, we are still missing studies that rigorously compare multiple state-of-the-art architectures on standard datasets with the exact same setup.

For this kind of work, the *Augmented Alzheimer MRI Dataset V2* is a great resource. It offers a four-class breakdown: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Augmenting the data definitely helps fix the issue of having too few scans, but it's a double-edged sword—if you aren't careful, it can easily lead to data leakage. That means we have to be incredibly strict about how the dataset is split and evaluated to make sure the model's performance is actually reliable.

Digging through all this literature, one thing stood out to us: even though there are tons of deep learning models out there for classifying Alzheimer's, we really lack a comprehensive, side-by-side comparison of the top architectures under one unified framework. Most studies look at a few models but rarely evaluate a standard CNN alongside VGG16, VGG19, DenseNet, Xception, and InceptionResNetV2 all at once on the same exact dataset.

Addressing this missing piece became the primary motivation for our research.Our goal with this study is to fill that specific gap by systematically testing and comparing these different architectures for multiclass Alzheimer's disease classification, using the MRI scans from the *Augmented Alzheimer MRI Dataset V2*.

Reff.

1. J. Wen et al., "Convolutional Neural Networks for Classification of Alzheimer's Disease: Overview and Reproducible Evaluation," https://arxiv.org/abs/1904.07773

2. S. Mohsen, “Alzheimer’s disease detection using deep learning and machine learning: a review,” Artificial Intelligence Review, vol. 58, art. 262, 2025. doi: 10.1007/s10462-025-11258-y https://link.springer.com/article/10.1007/s10462-025-11258-y

3. S. M. Mousavi, K. Moulaei, and L. Ahmadian, “Classifying and diagnosing Alzheimer’s disease with deep learning using 6735 brain MRI images,” Scientific Reports, vol. 15, art. 22721, 2025. doi: 10.1038/s41598-025-08092-1 https://www.nature.com/articles/s41598-025-08092-1

4. M. Alruily et al., “Ensemble deep learning for Alzheimer’s disease diagnosis using MRI: Integrating features from VGG16, MobileNet, and InceptionResNetV2 models,” PLoS ONE, vol. 20, no. 4, e0318620, 2025. doi 0.1371/journal.pone.0318620

https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0318620&utm

5. A. M. El-Assy et al., “A novel CNN architecture for accurate early detection and classification of Alzheimer’s disease using MRI data,” *Scientific Reports*, vol. 14, art. 3463, 2024. doi:10.1038/s41598-024-53733-6 https://www.nature.com/articles/s41598-024-53733-6

6. https://link.springer.com/article/10.1186/s40537-022-00650-y

7. Leveraging Deep Learning and Xception Architecture for High-Accuracy MRI Classification in Alzheimer Diagnosis. https://arxiv.org/abs/2403.16212

8. RETRACTED: Early detection and classification of Alzheimer’s disease through data fusion of MRI and DTI images using the YOLOv11 neural network

https://www.frontiersin.org/articles/10.3389/fnins.2025.1554015/full

9. An efficient method for early Alzheimer’s disease detection based on MRI images using deep convolutional neural networks

https://doi.org/10.3389/frai.2025.1563016

10. Alzheimer's Disease Detection Using Deep Learning on Neuroimaging: A Systematic Review

https://doi.org/10.3390/make6010024

11. Lightweight Deep Learning Models with Explainable AI for Early Alzheimer's Detection from Standard MRI Scans

https://doi.org/10.3390/diagnostics15212709

datasets: Kaggle, "Augmented Alzheimer MRI Dataset V2," Kaggle Dataset. https://www.kaggle.com/datasets/uraninjo/augmented-alzheimer-mri-dataset-v2